DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

PH1020 Physics II
Problem Set 4

Electric Fields in Matter Date: 25 Feb 2025

1. Prove the following identity for any vector field $\mathbf{P}(\mathbf{r})$:

$$\int_V dV \mathbf{P}(\mathbf{r}) = \int_S \mathbf{r}[\mathbf{P}(\mathbf{r}) \cdot \mathbf{dS}] - \int_V dV \mathbf{r}[\nabla \cdot \mathbf{P}(\mathbf{r})],$$

where V is the volume enclosed by the surface S.
Hint: Take the dot product of the above identity with a constant vector $\mathbf{a}$ to obtain a new identity. Prove this new identity and then argue that it implies the required identity.

2. A point dipole of moment $\mathbf{p} = p_0(\hat{x} + 2\hat{y} + 3\hat{z})$ is placed in an electrostatic potential ϕ given by

$$\phi(x, y, z) = \phi_0 \left[1 + \frac{(x^2 + y^2 + z^2)}{a^2} + \frac{(x^4 + y^4 + z^4)}{a^4}\right]$$

where ϕ_0 and a are appropriate constants. Find the force and the couple acting on the dipole when it is located at the point (a, a, a). Find also the torque of the force about the origin.

3. Starting from the equations for the polarization charge density and the polarization surface charge density, show that the net polarization charge in the volume and on the surface of a dielectric body is always zero.

4. Consider a dielectric medium with uniform polarisation $\boldsymbol{P}_0$. A spherical cavity is removed from inside this medium.
   (a) Find the bound surface charge density on the surface of the cavity.
   (b) Find the electric field at the center of the cavity due to the above surface charge.
   (c) Find the *total* electric field within the cavity, if the electric field in the bulk of the material is $\boldsymbol{E_0}$.
   (d) Find the displacement vector $\boldsymbol{D}$ everywhere.
   (e) Verify that the normal component of $\boldsymbol{D}$ satisfies the correct boundary condition at the surface of the cavity.

5. Consider a cylinder of length $2L$ and radius a, centred at the origin, and with it's symmetry axis being the z-axis. The cylinder carries a uniform polarisation $\boldsymbol{P} = P_0\hat{\boldsymbol{z}}$, with $P_0 =$ constant.
   (a) Find the bound charge densities ρ_b and σ_b.
   (b) Find the electric field on the positive z-axis. Check that it satisfies appropriate boundary condition at $z = L$.
   (c) Sketch the magnitude of the electric field at the origin as a function of a/L.

6. Consider a spherical capacitor with inner radius r_a and outer radius r_c. The total charge at the outer surface ($r = r_c$) is $-Q$ and at the inner surface ($r = r_a$) is Q. The space between these two surfaces is filled with a linear dielectric material with dielectric constant ϵ_{r_1} out to radius $r_b, r_a < r_b < r_c$, and another material with dielectric constant ϵ_{r_2} from r_b to r_c.

   (a) Find the displacement vector $\boldsymbol{D}$ everywhere.

   (b) Calculate the total bound charges on each of the boundaries. Hence verify the boundary conditions for normal component of the *electric field* at the interface of the two dielectrics.

   (c) Find the total capacitance of this system.

   (d) Repeat problem with $\epsilon_r = 1 + \alpha\left(\frac{x}{d}\right)$.

7. A sphere of radius R and dielectric constant κ, centred at the origin of coordinates, is placed in a constant electric field $\mathbf{E}_0$ directed along the z axis. The corresponding electrostatic potential is given by $\phi(r, \theta, \varphi) = (-E_0 r + b_1 r^{-2}) \cos\theta$ outside the sphere, and $\phi(r, \theta, \varphi) = b_2 r \cos\theta$ inside the sphere. Find

   (a) the constants b_1 and b_2, in terms of κ, E_0 and R;

   (b) the electric field at all points in space;

   (c) the polarization $\mathbf{P}$ of the sphere, and the dipole moment of the sphere about the origin;

   (d) the volume and surface densities of the bound charge in the sphere.

8. A cylindrical coaxial cable, lying along the z-axis has conducting surfaces at $\rho = a$ and $\rho = 4a$, which carry uniform surface charge densities σ_0 and $-\sigma_0/4$, respectively. The region between the two conductors is filled with linear dielectric media with dielectric constants K_1 and K_2 in the regions $a < \rho \leq 2a$ and $2a < \rho < 4a$, respectively.

   (a) Find the energy density between $a < \rho \leq 2a$.

   (b) Determine the ratio of the magnitude of the polarization just inside and just outside the boundary at $\rho = 2a$.

   (c) Sketch $|E|$ as a function of ρ in the interval $0 < \rho \leq 5a$ when $K_1 = 1.5$ and $K_2 = 2$.

1. Prove the following identity for any vector field $\mathbf{P}(\mathbf{r})$:

$$\int_V dV\mathbf{P}(\mathbf{r}) = \int_S \mathbf{r}[\mathbf{P}(\mathbf{r}) \cdot \mathbf{dS}] - \int_V dV\mathbf{r}[\nabla \cdot \mathbf{P}(\mathbf{r})],$$

where V is the volume enclosed by the surface S.
Hint: Take the dot product of the above identity with a constant vector $\mathbf{a}$ to obtain a new identity. Prove this new identity and then argue that it implies the required identity.

$$\vec{a} \cdot \int \vec{P}(\vec{r})\, dV = \int_S (\vec{a}\cdot\vec{r})(\vec{P}(\vec{r})\cdot d\vec{S}) - \int_V dV\,(\vec{a}\cdot\vec{r})[\vec{\nabla}\cdot P(\vec{r})]$$

$$\frac{d(uv)}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$(\vec{a}\cdot\vec{r})\,\vec{\nabla}\cdot\vec{P} = \vec{\nabla}(u\vec{v}) - (\vec{\nabla}u)\cdot\vec{v}$$

$$= \vec{\nabla}[(\vec{a}\cdot\vec{r})\vec{P}] - (\vec{\nabla}(\vec{a}\cdot\vec{r}))\cdot\vec{P}$$

$$\Rightarrow \int (\vec{a}\cdot\vec{r})\,\vec{\nabla}\cdot\vec{P}\, dV = \int \vec{\nabla}[(\vec{a}\cdot\vec{r})\vec{P}]\, dV - \int (\vec{\nabla}(\vec{a}\cdot\vec{r}))\,\vec{P}\, dV$$

$$\Rightarrow \int (\vec{a}\cdot\vec{r})\,\vec{\nabla}\cdot\vec{P}\, dV = \int_S (\vec{a}\cdot\vec{r})\,\vec{P}\cdot d\vec{S} - \int (\vec{\nabla}(\vec{a}\cdot\vec{r}))\,\vec{P}\, dV$$

gauss divergence theorm

$$\int [\vec{a}\cdot(\vec{\nabla}\vec{r})]\cdot\vec{P}\, dV = \int \vec{a}\cdot\vec{P}\, dV$$

$$\Rightarrow \int (\vec{a}\cdot\vec{r})(\vec{\nabla}\cdot\vec{P})\, dV = \int_S (\vec{a}\cdot\vec{r})(\vec{P}\cdot d\vec{S}) - \int \vec{a}\cdot\vec{P}\, dV$$

$$\Rightarrow \int \vec{r}(\vec{\nabla}\cdot\vec{P})\, dV = \int_S \vec{r}\,(\vec{P}\cdot d\vec{S}) - \int \vec{P}\, dV$$

2. A point dipole of moment $\mathbf{p} = p_0(\hat{x} + 2\hat{y} + 3\hat{z})$ is placed in an electrostatic potential ϕ given by

$$\phi(x, y, z) = \phi_0 \left[1 + \frac{(x^2 + y^2 + z^2)}{a^2} + \frac{(x^4 + y^4 + z^4)}{a^4}\right]$$

where ϕ_0 and a are appropriate constants. Find the force and the couple acting on the dipole when it is located at the point (a, a, a). Find also the torque of the force about the origin.

$$E = -\nabla\phi = -\phi_0\left(\frac{2x}{a^2} + \frac{4x^3}{a^4}\right)\hat{x} - \phi_0\left(\frac{2y}{a^2} + \frac{4y^3}{a^4}\right)\hat{y} - \phi_0\left(\frac{2z}{a^2} + \frac{4z^3}{a^4}\right)\hat{z}$$

$$\vec{E}(a,a,a) = \frac{-6\phi}{a}[\hat{x} + \hat{y} + \hat{z}]$$

$$\tau = \vec{p} \times \vec{E} = \frac{-6p_0\phi}{a}[-\hat{x} + 2\hat{y} - \hat{z}]$$

$$\vec{F} = (\vec{p}\cdot\nabla)\vec{E} = \frac{-p_0\phi_0}{a^2}\left[\left(2 + \frac{12x^2}{a^2}\right)\hat{x} + \left(2 + \frac{12y^2}{a^2}\right)\hat{y} + 3\left(2 + \frac{12z^2}{a^2}\right)\hat{z}\right]$$

$$\vec{F}(a,a,a) = \frac{-p_0\phi_0}{a^2}[14\hat{x} + 28\hat{y} + 42\hat{z}] = \frac{-14p_0\phi_0}{a^2}[\hat{x} + 2\hat{y} + 3\hat{z}]$$

$$\tau_{origin} = \vec{r} \times \vec{F} + \vec{\tau}$$

$$= \frac{-14p_0\phi_0}{a}[\hat{x} - 2\hat{y} + \hat{z}] - \frac{6p_0\phi_0}{a}[-\hat{x} + 2\hat{y} - \hat{z}]$$

$$= \frac{-8p_0\phi_0}{a}[\hat{x} - 2\hat{y} + \hat{z}]$$

3. Starting from the equations for the polarization charge density and the polarization surface charge density, show that the net polarization charge in the volume and on the surface of a dielectric body is always zero.

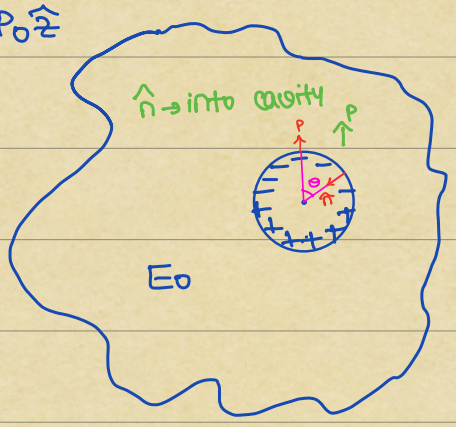
$\sigma_b = \vec{P}\cdot\hat{n}$ $\quad \rho_b = -\vec{\nabla}\cdot\vec{P}$

material is electrically neutral. Charges move inside the material

$$\oint_S (\vec{P}\cdot\hat{n})\,da - \int_V (\vec{\nabla}\cdot\vec{P})\,dV$$

$$= \oint_S \vec{P}\cdot d\vec{a} - \int (\vec{\nabla}\cdot\vec{P})\,dV = 0 \quad \text{(By gauss divergence th)}$$

4. Consider a dielectric medium with uniform polarisation $\boldsymbol{P}_0$. A spherical cavity is removed from inside this medium.
   (a) Find the bound surface charge density on the surface of the cavity.
   (b) Find the electric field at the center of the cavity due to the above surface charge. $\frac{1}{4\pi\varepsilon_0}\int \sigma_b\, dA/r^2\,(-\hat{r})$
   (c) Find the *total* electric field within the cavity, if the electric field in the bulk of the material is $\boldsymbol{E}_0$.
   (d) Find the displacement vector $\boldsymbol{D}$ everywhere.
   (e) Verify that the normal component of $\boldsymbol{D}$ satisfies the correct boundary condition at the surface of the cavity.



(a) $\sigma_b = \vec{P}\cdot\hat{n} = -P_0\cos\theta$

(b) $\frac{1}{4\pi\varepsilon_0}\int \frac{-P_0\cos\theta\cdot r^2\sin\theta\,d\theta\,d\phi}{r^2}\,(-\sin\theta\cos\phi\,\hat{x} - \sin\theta\sin\phi\,\hat{y} - \sin\phi\,\hat{z})$

$$\boxed{E = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma_b\,dA}{|r^2|}\,(-\hat{r})} = \frac{P_0}{3\varepsilon_0}\hat{z}$$

(c) $\vec{E_0} + \frac{\vec{P_0}}{3\varepsilon_0}$ (as if no cavity - due to cavity)

(d) $D_{\text{outside cavity}} = \varepsilon_0\bar{E}_0 + \bar{P}_0$

$D_{\text{inside}} = \varepsilon_0\bar{E} = \varepsilon_0\bar{E}_0 + \frac{\bar{P}_0}{3}$

(e) $E_{\text{out}}^{\perp} - E_{\text{in}}^{\perp} = \frac{\sigma_f}{\varepsilon_0} \Rightarrow \left[\bar{E}_0 - \left(E_0 + \frac{P_0}{3\varepsilon_0}\right)\right]\cos\theta = \frac{-P_0\cos\theta + \sigma_f}{\varepsilon_0}$

$\Rightarrow D_{\text{out}}^{\perp} - D_{\text{in}}^{\perp} = \varepsilon_0\bar{E} + \bar{P}_0 - \left[\varepsilon_0\bar{E} + \bar{P}_0/3\right] = \frac{2P_0\cos\theta}{3} = \sigma_f$

5. Consider a cylinder of length $2L$ and radius a, centred at the origin, and with it's symmetry axis being the z-axis. The cylinder carries a uniform polarisation $\boldsymbol{P} = P_0\hat{\boldsymbol{z}}$, with $P_0 =$ constant.

   (a) Find the bound charge densities ρ_b and σ_b.

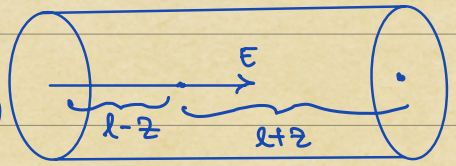
   (b) Find the electric field on the positive z-axis. Check that it satisfies appropriate boundary condition at $z = L$.

   (c) Sketch the magnitude of the electric field at the origin as a function of a/L.

(a) $\rho_b = 0$; $(\sigma_b)_{top} = P_0$, $(\sigma_b)_{bottom} = -P_0$, $(\sigma_b)_{curved} = 0$

(b) $(z < L \; ; \; z > -L)$

$$\frac{P_0}{2\varepsilon_0}\left[1 - \frac{(L-z)}{\sqrt{r^2+(L-z)^2}}\right](-\hat{z})$$



$$+\frac{P_0}{2\varepsilon_0}\left[1 - \frac{(L+z)}{\sqrt{r^2+(L+z)^2}}\right](-\hat{z})$$

$$z > L: \; \frac{P_0}{2\varepsilon_0}\left[1 - \frac{z-L}{\sqrt{r^2+(z-L)^2}} - 1 + \frac{z+l}{\sqrt{r^2+(z+l)^2}}\right]\hat{z}$$

$z < L$: similar

$$z = L^+ \Rightarrow E_+ = \frac{P_0}{2\varepsilon_0}\frac{2l}{\sqrt{r^2+4l^2}}$$

$$z = L^- \Rightarrow E_- = \frac{P_0}{2\varepsilon_0}\left[2 - \frac{2l}{\sqrt{r^2+4l^2}}\right] \Rightarrow E_+ - E_- = \frac{P_0}{\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

(c) Put $z = 0$

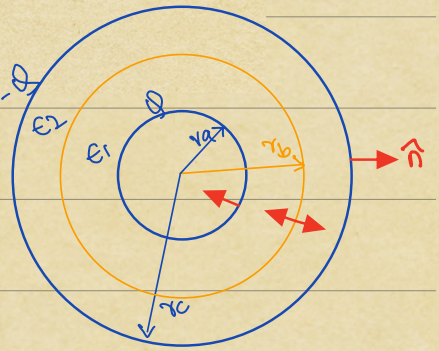
6. Consider a spherical capacitor with inner radius r_a and outer radius r_c. The total charge at the outer surface ($r = r_c$) is $-Q$ and at the inner surface ($r = r_a$) is Q. The space between these two surfaces is filled with a linear dielectric material with dielectric constant ϵ_{r_1} out to radius $r_b, r_a < r_b < r_c$, and another material with dielectric constant ϵ_{r_2} from r_b to r_c.

(a) Find the displacement vector $\boldsymbol{D}$ everywhere.

(b) Calculate the total bound charges on each of the boundaries. Hence verify the boundary conditions for normal component of the *electric field* at the interface of the two dielectrics.

(c) Find the total capacitance of this system.

(d) Repeat problem with $\epsilon_r = 1 + \alpha\left(\frac{x}{a}\right)$.



a) $r \le a \Rightarrow D = 0$

$r \in (a,c) \Rightarrow D = \dfrac{Q}{4\pi r^2}$

$r \ge c \Rightarrow D = 0 \quad (Q + (-Q) = 0)$

b) $\bar{D} = \epsilon_0 \epsilon_r \bar{E} = \epsilon_0 \bar{E} + \bar{P}$

$\bar{D} = \dfrac{\bar{D}}{\epsilon_r} + \bar{P} \Rightarrow \bar{P} = \bar{D}\left(1 - \dfrac{1}{\epsilon_r}\right)$

$$\vec{P} = \begin{cases} Q/4\pi r^2 (1 - 1/\epsilon_{r_1})\,\hat{r}; & r \in (a,b) \\ Q/4\pi r^2 (1 - 1/\epsilon_{r_2})\,\hat{r}; & r \in (b,c) \\ 0 & ; \text{else} \end{cases}$$

$$\sigma_{bound} = \begin{cases} -Q/4\pi r^2 (1 - 1/\epsilon_{r_1}) & r = r_a \\ Q/4\pi r^2 (1/\epsilon_{r_2} - 1/\epsilon_{r_1}) & r = r_b \\ Q/4\pi r^2 (1 - 1/\epsilon_{r_2}) & r = r_c \end{cases}$$

c) $$-\int_{r_a}^{r_b} \frac{Q}{4\pi r^2 \epsilon_0 \epsilon_{r_1}} \cdot dr + -\int_{r_b}^{r_c} \frac{-Q \cdot dr}{4\pi r^2 \epsilon_0 \epsilon_{r_2}}$$

7. A sphere of radius R and dielectric constant κ, centred at the origin of coordinates, is placed in a constant electric field $\mathbf{E}_0$ directed along the z axis. The corresponding electrostatic potential is given by $\phi(r,\theta,\varphi) = (-E_0 r + b_1 r^{-2})\cos\theta$ outside the sphere, and $\phi(r,\theta,\varphi) = b_2 r\cos\theta$ inside the sphere. Find

(a) the constants b_1 and b_2, in terms of κ, E_0 and R;

(b) the electric field at all points in space;

(c) the polarization $\mathbf{P}$ of the sphere, and the dipole moment of the sphere about the origin;

(d) the volume and surface densities of the bound charge in the sphere.

(a) $\boxed{-E_0R + \frac{b_1}{R^2} = b_2R}$ $\quad E_0 + b_2 = \frac{b_1}{R^3}$ $\quad K = \varepsilon_r$ $\quad \bar{D} = K\varepsilon_0\bar{E}$

$$\vec{E} = -\vec{\nabla}.V = -\frac{dV}{dr}\hat{r} - \frac{1}{r}\frac{dV}{d\theta}\hat{\theta} - \frac{1}{r\sin\theta}\frac{dV}{d\phi}\hat{\phi}$$

$$E_{inside} = -b_2\cos\theta\,\hat{r} + b_2\sin\theta\,\hat{\theta}$$

$$E_{outside} = \left(E_0\cos\theta - \frac{2b_1}{r^3}\cos\theta\right)\hat{r} + \left(-E_0 + \frac{b_1}{r^3}\right)\sin\theta\,\hat{\theta}$$

$$\varepsilon_0\varepsilon_r\frac{dV_{in}}{dr} = \varepsilon_0\frac{dV_{out}}{dr} \qquad D_{in}^{\perp} = D_{out}^{\perp}$$

$$\varepsilon_r.(-b_2\cos\theta) = E_0\cos\theta - \frac{2b_1\cos\theta}{r^3}$$

$$b_1 = \left(\frac{K-1}{K+2}\right)E_0R^3 \qquad b_2 = -\left(\frac{3}{K+2}\right)E_0 \qquad E_0 + Kb_2 = \frac{2b_1}{R^3}$$

(b) Slap b_1, b_2 in the field equation

(c) $r < R \Rightarrow P = \bar{D} - \varepsilon_0\bar{E} = \frac{3(K-1)}{K+2}\varepsilon_0E_0\hat{z}$

p (dipole Moment) $= P \times 4/3\,\pi R^3$

(d) $\sigma_b = \vec{P}.\hat{n}$ $\quad \rho_b = -\vec{\nabla}.\vec{P} = 0$

8. A cylindrical coaxial cable, lying along the z-axis has conducting surfaces at $\rho = a$ and $\rho = 4a$, which carry uniform surface charge densities σ_0 and $-\sigma_0/4$, respectively. The region between the two conductors is filled with linear dielectric media with dielectric constants K_1 and K_2 in the regions $a < \rho \leq 2a$ and $2a < \rho < 4a$, respectively.

(a) Find the energy density between $a < \rho \leq 2a$.

(b) Determine the ratio of the magnitude of the polarization just inside and just outside the boundary at $\rho = 2a$.

(c) Sketch $|E|$ as a function of ρ in the interval $0 < \rho \leq 5a$ when $K_1 = 1.5$ and $K_2 = 2$.

(a) $\int \vec{D}.d\vec{s} = Q_{free}$

$\Rightarrow D \times 2\pi r \ell = \sigma_0 \times 2\pi(a)\ell \Rightarrow D = \frac{\sigma_0 a}{r}$

$\Rightarrow D = \varepsilon_0 \varepsilon_r E \Rightarrow E = \frac{\sigma_0 a}{\varepsilon_0 K_1 r}$

$u = \frac{1}{2} \times \frac{\sigma_0^2 a^2}{\varepsilon_0 K_1 r^2}$

(b) $\vec{P} = \vec{D} - \varepsilon_0 \vec{E}$

$P_{in} = \frac{a\sigma_0}{r} - \frac{a\sigma_0}{K_1 r}$ $\quad P_{out} = \frac{a\sigma_0}{r} - \frac{a\sigma_0}{r K_2}$

$\frac{P_1}{P_2} = \frac{1 - 1/K_1}{1 - 1/K_2}$

(c) $\rho < a \Rightarrow E = 0$

$\rho \in (a, 2a) \Rightarrow E = \frac{a\sigma_0}{\varepsilon_0 K_1 \rho}$

$\rho \in (2a, 4a) \Rightarrow E = \frac{a\sigma_0}{\varepsilon_0 K_2 \rho}$

$\rho > 4a \Rightarrow E = 0$ (Q enclosed is 0)

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PH1020 Physics II
Problem Set 4 - Solutions

Electric Fields in Matter Date: 25 Feb 2025

1. Problem-1

Question: Prove the following identity for any vector field $\mathbf{P}(\mathbf{r})$:

$$\int_V dV \mathbf{P}(\mathbf{r}) = \int_S \mathbf{r}[\mathbf{P}(\mathbf{r}) \cdot \mathbf{dS}] - \int_V dV \mathbf{r}[\nabla \cdot \mathbf{P}(\mathbf{r})],$$

where V is the volume enclosed by the surface S.
Hint: Take the dot product of the above identity with a constant vector $\mathbf{a}$ to obtain a new identity. Prove this new identity and then argue that it implies the required identity.
Solution:

Let $\boldsymbol{a}$ be a constant vector. Consider

$$\boldsymbol{a} \cdot \int_V dV \boldsymbol{r}(\boldsymbol{\nabla} \cdot \boldsymbol{P})$$

This is equal to

$$\begin{aligned}
\boldsymbol{a} \cdot \int_V dV \boldsymbol{r}(\boldsymbol{\nabla} \cdot \boldsymbol{P}) &= \int_V dV (\boldsymbol{a} \cdot \boldsymbol{r})(\boldsymbol{\nabla} \cdot \boldsymbol{P}) \quad (\text{ as } \boldsymbol{a} \text{ is a constant }) \\
&= \int_V dV \boldsymbol{\nabla} \cdot [(\boldsymbol{a} \cdot \boldsymbol{r})\boldsymbol{P}] - \int dV [\boldsymbol{\nabla}(\boldsymbol{a} \cdot \boldsymbol{r})] \cdot \boldsymbol{P} \\
&= \int_S (\boldsymbol{a} \cdot \boldsymbol{r})(\boldsymbol{P} \cdot d\boldsymbol{S}) - \boldsymbol{a} \cdot \int_V dV \boldsymbol{P} \\
&= \boldsymbol{a} \cdot \left(\int_S \boldsymbol{r}(\boldsymbol{P} \cdot d\boldsymbol{S}) - \int_V dV \boldsymbol{P} \right)
\end{aligned}$$

This is true for all $\boldsymbol{a}$. So

$$\int_V dV \boldsymbol{r}(\boldsymbol{\nabla} \cdot P) = \int_S \boldsymbol{r}(P \cdot d\boldsymbol{S}) - \int_V dV \boldsymbol{P}$$

2. Problem-2

Question: A point dipole of moment $\mathbf{p} = p_0 (\hat{x} + 2\hat{y} + 3\hat{z})$ is placed in an electrostatic potential ϕ given by

$$\phi(x, y, z) = \phi_0 \left[1 + \frac{(x^2 + y^2 + z^2)}{a^2} + \frac{(x^4 + y^4 + z^4)}{a^4} \right]$$

where ϕ_0 and a are appropriate constants. Find the force and the couple acting on the dipole when it is located at the point (a, a, a). Find also the torque of the force about the origin.

Solution:

We are given the following electrostatic potential

$$\phi(x, y, z) = \phi_0 + \frac{\phi_0}{a^2}\left(x^2 + y^2 + z^2\right) + \frac{\phi_0}{a^4}\left(x^4 + y^4 + z^4\right)$$

whose electric field is

$$\begin{aligned}\boldsymbol{E} &= -\boldsymbol{\nabla}\phi \\ &= -\left(\frac{\phi_0}{a^2}\right)\left[\left(2x + \frac{4x^3}{a^2}\right)\hat{\mathbf{x}} + \left(2y + \frac{4y^3}{a^2}\right)\hat{\mathbf{y}} + \left(2z + \frac{4z^3}{a^2}\right)\hat{\mathbf{z}}\right].\end{aligned}$$

Using the standard formula for the force on a point dipole in a spatially inhomogenous electrostatic field, i.e., $\boldsymbol{F} = (\boldsymbol{p}\cdot\boldsymbol{\nabla})\boldsymbol{E}$.[1]

$$\begin{aligned}\boldsymbol{F} &= (\boldsymbol{p}\cdot\boldsymbol{\nabla})\boldsymbol{E}|_{(a,a,a)} \\ &= p_0\left(\frac{\partial}{\partial x} + 2\frac{\partial}{\partial y} + 3\frac{\partial}{\partial z}\right)\boldsymbol{E}\Big|_{(a,a,a)} \\ &= -\frac{p_0\phi_0}{a^2}\left[\left(2 + \frac{12x^2}{a^2}\right)\hat{\mathbf{x}} + 2\left(2 + \frac{12y^2}{a^2}\right)\hat{\mathbf{y}} + 3\left(2 + \frac{12z^2}{a^2}\right)\hat{\mathbf{y}}\right]\Bigg|_{(a,a,a)} \\ \boldsymbol{F} &= -\frac{14p_0\phi_0}{a^2}\left(\hat{\mathbf{x}} + 2\hat{\mathbf{y}} + 3\hat{\mathbf{z}}\right).\end{aligned}$$

The torque, $\boldsymbol{\tau}$, on a point dipole in an electric field is given by $\boldsymbol{\tau} = (\boldsymbol{p}\times\boldsymbol{E})$, where $\boldsymbol{E}$ is the electric field at the location of the point dipole. Computing

$$\begin{aligned}\boldsymbol{\tau} &= (\boldsymbol{p}\times\boldsymbol{E})|_{(a,a,a)} \\ \boldsymbol{\tau} &= \frac{6p_0\phi_0}{a}\left(\hat{\mathbf{x}} - 2\hat{\mathbf{y}} + \hat{\mathbf{z}}\right).\end{aligned}$$

The torque, $\boldsymbol{\tau}$, about the origin is

$$\boldsymbol{\tau}|_{(a,a,a)} + a\left(\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}\right)\times\boldsymbol{F} = -\frac{8p_0\phi_0}{a}\left(\hat{\mathbf{x}} - 2\hat{\mathbf{y}} + \hat{\mathbf{z}}\right).$$

3. Problem-3

Question: Starting from the equations for the polarization charge density and the polarization surface charge density, show that the net polarization charge in the volume and on the surface of a dielectric body is always zero.

Solution:

[01] This is easily derived using the construction of a point dipole as a limit of two charges with equal in magnitude but with opposite signs.

Consider a surface S that bounds a volume V. Due to the Polarization, $\mathbf{P}$, there is a surface charge density $\sigma_p = \mathbf{P}.\hat{n}$, and a volume charge density given by $\rho_p = -\nabla \cdot \mathbf{P}$. The net polarization charge is

$$\int_{\mathrm{S}} \sigma_p dS + \int_{\mathrm{V}} \rho_p dV$$
$$= \int_{\mathrm{S}} \mathbf{P} \cdot \hat{n} dS + \int_{\mathrm{V}} -\nabla \cdot \mathbf{P} dv = 0$$

wherein in the last line we use Gauss's divergence theorem.

4. Problem-4

Question: Consider a dielectric medium with uniform polarisation $\boldsymbol{P}_0$. A spherical cavity is removed from inside this medium.

(a) Find the bound surface charge density on the surface of the cavity.

(b) Find the electric field at the center of the cavity due to the above surface charge.

(c) Find the *total* electric field within the cavity, if the electric field in the bulk of the material is $\boldsymbol{E_0}$.

(d) Find the displacement vector $\boldsymbol{D}$ everywhere.

(e) Verify that the normal component of $\boldsymbol{D}$ satisfies the correct boundary condition at the surface of the cavity.

Solution:

$\frac{1}{4\pi\epsilon_0}\int \frac{\sigma dA}{R^2}$ $dA = R^2 \sin\theta \, d\theta \, d\phi \, (-\hat{r})$

$= \frac{1}{4\pi\epsilon_0}\int -P_0 \cos\theta \sin\theta \, d\theta \, d\phi$

$[-\sin\theta\cos\phi\,\hat{x} - \sin\theta\sin\phi\,\hat{y} - \cos\theta\,\hat{z}]$

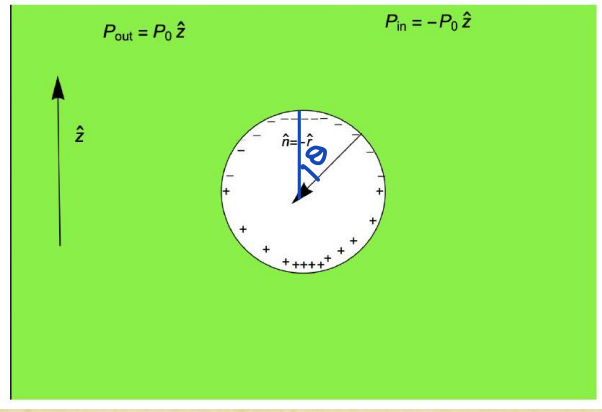


Figure 1: The green colored area is the bulk material and the white region is the cavity

(a) It is given that polarization is $\mathbf{P}_0$. Without loss of generality we take the z axis along the direction of polarization such that $\mathbf{P}_0 = |\mathbf{P}_0|\,\hat{\mathbf{z}} = P_0\hat{\mathbf{z}}$. Then, the bound surface charge density

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = -\,|\mathbf{P}_0| \cos\theta$$

"θ" is the usual spherical polar angle. Here, we have expressed the bound charge in terms of the polarization of the bulk material $\mathbf{P}_0$. Observe here that the unit normal vector $\hat{\mathbf{n}}$

is radially inward.

(b) The electric field at the center of the cavity due to the above surface charge density

$$\begin{aligned}
\mathbf{E} &= \frac{1}{4\pi\epsilon_0}\int \frac{\sigma_b dA}{|\mathbf{r}|^2}(-\hat{\mathbf{r}}) \\
&= \frac{1}{4\pi\epsilon_0}\int_{\theta=0}^{\pi} d\theta \int_{\phi=0}^{2\pi} d\phi \frac{(-|\mathbf{P}_0|\cos\theta)\,|\mathbf{r}|^2 \sin\theta}{|\mathbf{r}|^2}(-\sin\theta\cos\phi\hat{\mathbf{x}} - \sin\theta\sin\phi\hat{\mathbf{y}} - \cos\theta\hat{\mathbf{z}}) \\
&= \frac{|\mathbf{P}_0|}{4\pi\epsilon_0}\int_{\theta=0}^{\pi} d\theta \int_{\phi=0}^{2\pi} d\phi \left(\hat{\mathbf{x}}\left(\sin^2\theta\cos\theta\cos\phi\right) + \hat{\mathbf{y}}\left(\sin^2\theta\cos\theta\sin\phi\right) + \hat{\mathbf{z}}\left(\cos^2\theta\sin\theta\right)\right) \\
&= \frac{|\mathbf{P}_0|}{4\pi\epsilon_0}\int_{\theta=0}^{\pi} d\theta \int_{\phi=0}^{2\pi} d\phi \cos^2\theta\sin\theta\hat{\mathbf{z}} \\
&= \frac{\mathbf{P}_0}{4\pi\epsilon_0} \times 2\pi \times \frac{2}{3} \\
&= \frac{\mathbf{P}_0}{3\epsilon_0}
\end{aligned}$$

(c) In order to find the total electric field at the center of the cavity, we can consider the cavity as superposition of two systems: (i) The whole dielectric as if there is no cavity and then (ii) a sphere with exactly negative polarization and as exact size of the cavity. Hence,

$$\mathbf{E}_{\text{total}} = \mathbf{E}_0 + \frac{\mathbf{P}_0}{3\epsilon_0}$$

(d) The subscript "out" means outside cavity and "in" means inside cavity.

$$\begin{aligned}
\mathbf{D}_{\text{out}} &= \epsilon_0\mathbf{E}_0 + \mathbf{P}_0 \\
\mathbf{D}_{\text{in}} &= \epsilon_0\mathbf{E} = \epsilon_0\left(\mathbf{E}_0 + \frac{\mathbf{P}_0}{3\epsilon_0}\right) = \epsilon_0\mathbf{E}_0 + \frac{\mathbf{P}_0}{3}
\end{aligned}$$

(e)

$$\begin{aligned}
E^{\perp}_{\text{out}} - E^{\perp}_{\text{in}} &= \frac{\sigma}{\epsilon_0} \\
-\frac{P_0\cos\theta}{3\epsilon_0} &= \frac{\sigma_b + \sigma_f}{\epsilon_0} \\
\sigma_f &= \frac{2P_0\cos\theta}{3}
\end{aligned}$$

Now.

$$D^{\perp}_{\text{out}} - D^{\perp}_{\text{in}} = P_0\cos\theta - \frac{P_0\cos\theta}{3} = \frac{2P_0\cos\theta}{3} = \sigma_f$$

This is consistent with equation (4.26) of Griffith's book.

5. Problem-5

Question: Consider a cylinder of length $2L$ and radius a, centred at the origin, and with it's symmetry axis being the z-axis. The cylinder carries a uniform polarisation $\boldsymbol{P} = P_0\hat{\boldsymbol{z}}$, with P_0 = constant.

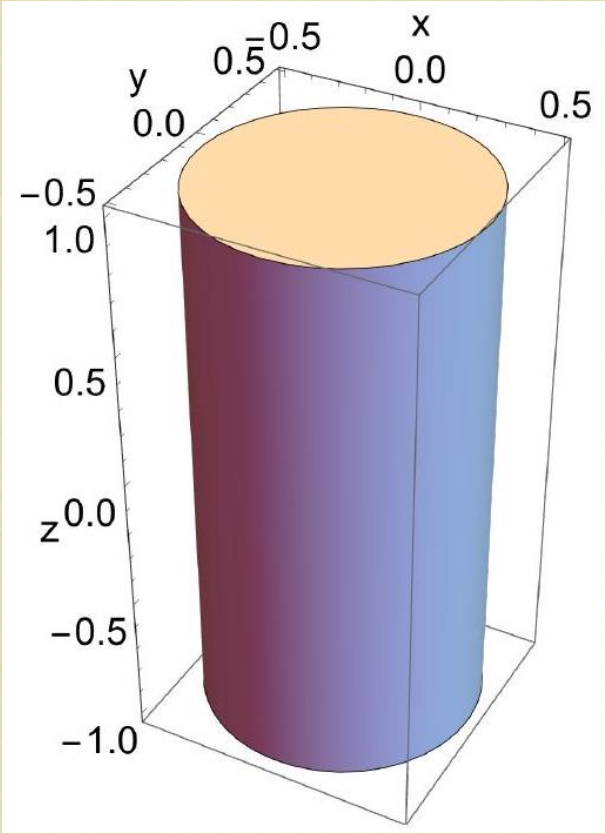


Figure 2: The cylinder whose symmetry axis is along $\hat{\mathbf{z}}$ and in units of L.

(a) Find the bound charge densities ρ_b and σ_b.

(b) Find the electric field on the positive z-axis. Check that it satisfies appropriate boundary condition at $z = L$.

(c) Sketch the magnitude of the electric field at the origin as a function of a/L.

Solution:

(a) The cylinder runs from $-L$ to L along the z axis being the symmetric axis of the cylinder, see Figure 2.

$$\rho_b = -\boldsymbol{\nabla} \cdot \mathbf{P} = 0$$

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = \begin{cases} +P_0 & \text{on the top circular surface at } z = +L \\ -P_0 & \text{on the bottom circular surface at } z = -L \\ 0 & \text{on the curved surface} \end{cases}$$

(b) The electric field at a point z above the origin,

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_{\text{due to plate at } z=L} + \mathbf{E}_{\text{due to plate at } z=-L} \\ &= \left[\frac{P_0}{2\epsilon} \left(1 - \frac{L-z}{\sqrt{a^2 + (L-z)^2}} \right) + \frac{P_0}{2\epsilon} \left(1 - \frac{L+z}{\sqrt{a^2 + (L+z)^2}} \right) \right] (-\hat{\mathbf{z}}) \quad \text{for} \quad 0 < z < L \\ &= \frac{P_0}{2\epsilon} \left(2 - \frac{L-z}{\sqrt{a^2 + (L-z)^2}} - \frac{L+z}{\sqrt{a^2 + (L+z)^2}} \right) (-\hat{\mathbf{z}}) \quad \text{for} \quad 0 < z < L \end{aligned}$$

Verifying the boundary condition at $z = L$: As there is no source, therefore, $\sigma_f = 0$. Thus, verifying the boundary condition for the normal component of $\mathbf{D}$,

$$\begin{aligned} D^{\perp}_{\text{above}} - D^{\perp}_{\text{below}} &= \left(\epsilon_0 E^{\perp}_{\text{above}}\right) - \left(\epsilon_0 E^{\perp}_{\text{below}} + P^{\perp}_0\right) \\ &= \epsilon_0 \left(E^{\perp}_{\text{above}} - E^{\perp}_{\text{below}}\right) - P^{\perp}_0 \\ &= \epsilon_0 \tfrac{\sigma}{\epsilon_0} - P_0 \\ &= \sigma_b - P_0 \qquad\qquad (\sigma = \sigma_b + \sigma_f \quad \text{but } \sigma_f = 0) \\ &= P_0 - P_0 \end{aligned}$$

(c) Case I: $\frac{a}{L} \to 0$: Substituting $a/L \to 0$ in the above expression of $\mathbf{E}$ we see the electric field vanishes; which of course make sense because when the radius of the cylinder is much much smaller compared to the length of it then, the observer at the origin sees the charge distribution at $\pm L$ as point charges. Hence the whole system behaves like a dipole. See Figure 3 Case II : $\frac{a}{L} \to \infty$: In this case

$$\mathbf{E} = \frac{P_0}{\epsilon_0} = \frac{\text{surface charge density}}{\epsilon_0}$$

This is exactly the field inside a parallel plate capacitor which can be intuitively seen from the geometry. See Figure 3

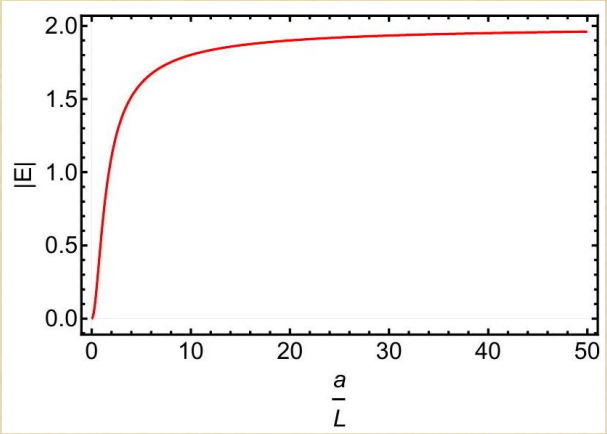


Figure 3: This is the plot of magnitude of electric field of the configuration vs a/L. The y axis is in the units of $P_0/2\epsilon_0$. One can clearly see the field vanishes at the origin which indicates the observer at the origin sees the charge distribution behaves as point charges. Similarly, at large a/L the curve approaches 2 , which implies the system behaves as a parallel plate capacitor.

6. Problem-6

Question: Consider a spherical capacitor with inner radius r_a and outer radius r_c. The total charge at the outer surface ($r = r_c$) is $-Q$ and at the inner surface ($r = r_a$) is Q. The space between these two surfaces is filled with a linear dielectric material with dielectric constant ϵ_{r_1} out to radius $r_b, r_a < r_b < r_c$, and another material with dielectric constant ϵ_{r_2} from r_b to r_c.

(a) Find the displacement vector $\boldsymbol{D}$ everywhere.

(b) Calculate the total bound charges on each of the boundaries. Hence verify the boundary conditions for normal component of the *electric field* at the interface of the two dielectrics.

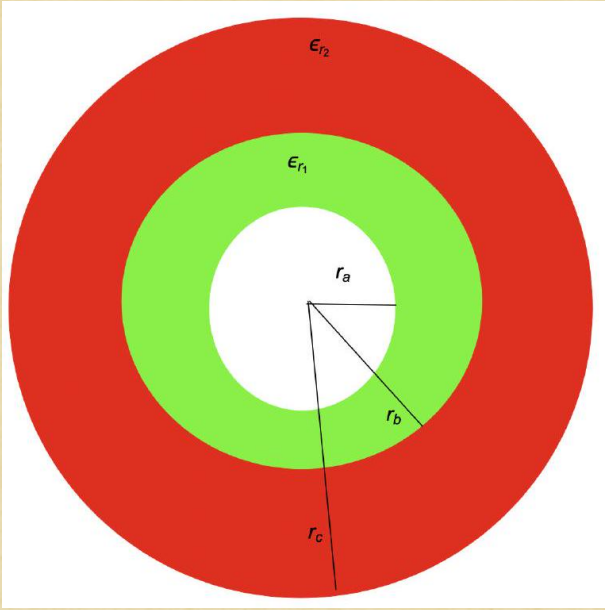
(c) Find the total capacitance of this system.

(d) Repeat problem with $\epsilon_r = 1 + \alpha\left(\frac{x}{d}\right)$.

Solution:

(a) There is $+Q$ charge on the innermost surface i.e. at $r = r_a$ and $-Q$ at the outermost surface i.e. at $r = r_c$. i. Using the Gauss' law for the displacement vector $\mathbf{D}$ we get

$$\mathbf{D} = \begin{cases} \frac{Q}{4\pi r^2}\hat{\mathbf{r}} & r_a < r < r_c \\ \mathbf{0} & \text{elsewhere} \end{cases}$$



Hence the electric field, ($\mathbf{D} = \epsilon\mathbf{E} = \epsilon_0\epsilon_r\mathbf{E}$)

$$\mathbf{E} = \begin{cases} \frac{1}{4\pi\epsilon_0\epsilon_{r_1}}\frac{Q}{r^2}\hat{\mathbf{r}} & r_a < r < r_b \\ \frac{1}{4\pi\epsilon_0\epsilon_{r_2}}\frac{Q}{r^2}\hat{\mathbf{r}} & r_b < r < r_c \\ \mathbf{0} & \text{elsewhere} \end{cases}$$

(b) To compute the bound charges we need the polarization first. Using $\mathbf{P} = \mathbf{D} - \epsilon_0\mathbf{E}$,

$$\mathbf{P} = \begin{cases} \frac{Q}{4\pi r^2}\left(1 - \frac{1}{\epsilon_{r_1}}\right)\hat{\mathbf{r}} & r_a < r < r_b \\ \frac{Q}{4\pi r^2}\left(1 - \frac{1}{\epsilon_{r_2}}\right)\hat{\mathbf{r}} & r_b < r < r_c \end{cases}$$

Therefore,

$$\sigma_b = \begin{cases} -\frac{Q}{4\pi r_a^2}\left(1-\frac{1}{\epsilon_{r_1}}\right) & \text{at} \quad r = r_a \\ \frac{Q}{4\pi r_b^2}\left(\frac{1}{\epsilon_{r_2}}-\frac{1}{\epsilon_{r_1}}\right) & \text{at} \quad r = r_b \\ \frac{Q}{4\pi r_c^2}\left(1-\frac{1}{\epsilon_{r_2}}\right) & \text{at} \quad r = r_c \end{cases}$$

Following the boundary condition (4.40) of Griffith's book

$$\epsilon_{\text{above}}\, E^{\perp}_{\text{above}} - \epsilon_{\text{below}}\, E^{\perp}_{\text{below}} = \begin{cases} \frac{Q}{4\pi r^2} = \sigma_f & \text{at} \quad r = r_a \\ 0 & \text{at} \quad r = r_b \\ -\frac{Q}{4\pi r^2} = \sigma_f & \text{at} \quad r = r_c \end{cases}$$

(c) The potential difference between the plates,

$$\begin{aligned} V &= \frac{Q}{4\pi\epsilon_0}\left(\int_{r_a}^{r_b}\frac{1}{\epsilon_{r_1} r^2}dr + \int_{r_b}^{r_c}\frac{1}{\epsilon_{r_2} r^2}dr\right) \\ &= \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{\epsilon_{r_1}}\left(\frac{1}{r_a}-\frac{1}{r_b}\right)+\frac{1}{\epsilon_{r_2}}\left(\frac{1}{r_b}-\frac{1}{r_c}\right)\right) \end{aligned}$$

Capacitance,

$$C = \frac{Q}{V} = \frac{4\pi\epsilon_0}{\left(\frac{1}{\epsilon_{r_1}}\left(\frac{1}{r_a}-\frac{1}{r_b}\right)+\frac{1}{\epsilon_{r_2}}\left(\frac{1}{r_b}-\frac{1}{r_c}\right)\right)}$$

(d) Using the Gauss' law for $\mathbf{D}$, the displacement vector

$$\mathbf{D} = \sigma\hat{\mathbf{x}}$$

The electric field

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon} = \frac{\sigma}{\epsilon}\hat{\mathbf{x}} = \frac{\sigma}{\epsilon_0\epsilon_r}\hat{\mathbf{x}} = \frac{\sigma}{\epsilon_0\left(1+\alpha\frac{x}{d}\right)}\hat{\mathbf{x}}$$

The polarization vector

$$\mathbf{P} = \mathbf{D} - \epsilon_0\mathbf{E} = \sigma\left(1-\frac{1}{\epsilon_r}\right)\hat{\mathbf{x}}$$

The bound surface charge density,

$$\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}} = \begin{cases} 0 & \text{at} \quad x = 0 \\ \sigma\frac{\alpha}{1+\alpha} & \text{at} \quad x = d \end{cases}$$

The bound volume charge density

$$\rho_b = -\boldsymbol{\nabla}\cdot\mathbf{P} = -\sigma\frac{\alpha}{d}\left(1+\frac{\alpha x}{d}\right)^{-2}$$

The potential difference between the plates,

$$\begin{aligned} V &= \int_0^d \mathbf{E}\cdot\mathbf{dx} \\ &= \int_0^d \frac{\sigma}{\epsilon_0}\frac{1}{\left(1+\frac{\alpha x}{d}\right)}dx \\ &= \frac{\sigma d}{\epsilon_0}\int_0^1 \frac{1}{1+\alpha y}dy \quad \left(y=\frac{x}{d}\right) \\ &= \frac{\sigma d}{\epsilon_0}\frac{\ln(1+\alpha)}{\alpha} \end{aligned}$$

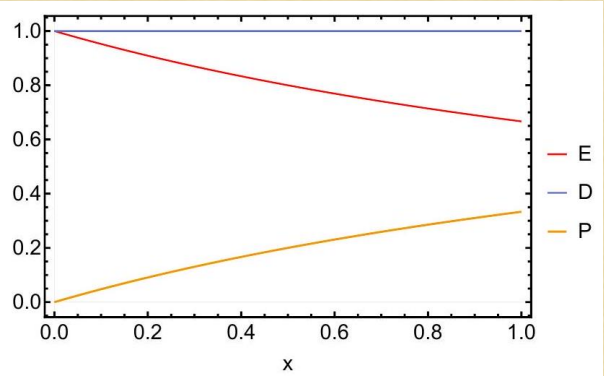


Figure 4: Plot of the magnitude of $\mathbf{E}, \mathbf{D}, \mathbf{P}$ in the units of σ and x is in the units of d. We have taken $\epsilon_0 = \alpha = 1$

The capacitance,

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d} \frac{1}{\frac{\ln(1+\alpha)}{\alpha}}$$

When $\alpha \to 0$, $\frac{\ln(1+\alpha)}{\alpha} \approx 1 - \mathcal{O}(\alpha)$, hence $C = \frac{\epsilon_0 A}{d}$.

7. Problem-7

Question: A sphere of radius R and dielectric constant κ, centred at the origin of coordinates, is placed in a constant electric field $\mathbf{E}_0$ directed along the z axis. The corresponding electrostatic potential is given by $\phi(r, \theta, \varphi) = (-E_0 r + b_1 r^{-2}) \cos\theta$ outside the sphere, and $\phi(r, \theta, \varphi) = b_2 r \cos\theta$ inside the sphere. Find

(a) the constants b_1 and b_2, in terms of κ, E_0 and R;

(b) the electric field at all points in space;

(c) the polarization $\mathbf{P}$ of the sphere, and the dipole moment of the sphere about the origin;

(d) the volume and surface densities of the bound charge in the sphere.

Solution:

The electrostatic potential is given to be

$$\phi(r, \theta, \varphi) = \begin{cases} (-E_0 r + b_1 r^{-2}) \cos\theta, & \text{for } r > R \\ b_2 r \cos\theta, & \text{for } r < R \end{cases}$$

The corresponding electric field is given by

$$\mathbf{E} = \begin{cases} E_0 \hat{\mathbf{z}} + \frac{b_1}{r^3} \left(2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\theta} \right), & \text{for } r > R \\ -b_2 \hat{\mathbf{z}}, & \text{for } r < R \end{cases}$$

The displacement $\mathbf{D}$ can be computed using $\mathbf{D} = \epsilon_0 \kappa \mathbf{E}$ for $r < R$ and $\mathbf{D} = \epsilon_0 \mathbf{E}$ for $r > R$. We obtain

$$\frac{\mathbf{D}}{\epsilon_0} = \begin{cases} E_0 \hat{\mathbf{z}} + \dfrac{b_1}{r^3}\left(2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\theta}\right), & \text{for } r > R \\ -b_2 \kappa \hat{\mathbf{z}}, & \text{for } r < R \end{cases}$$

(a) In order to fix the constants b_1 and b_2 we use the following matching conditions at $r = R$ and arbitrary (θ, φ). First, we expect the normal component of $\mathbf{D}$, i.e., $\mathbf{D} \cdot \hat{\mathbf{r}}$ to be continuous as there is no free charge on the interface. Second, the tangential component of $\mathbf{E}$, i.e., $\mathbf{D} \times \hat{\mathbf{r}}$ should be continuous. We obtain (using $\hat{\mathbf{r}} \times \hat{\mathbf{z}} = -\sin\theta \hat{\varphi}$)

$$\left(E_0 + 2\frac{b_1}{R^3}\right)\cos\theta = -b_2 \kappa \cos\theta$$
$$\left(-E_0 + \frac{b_1}{R^3}\right)\sin\theta \hat{e}_\varphi = b_2 \sin\theta \hat{e}_\varphi$$

which can be solved to obtain

$$b_1 = \left(\frac{\kappa - 1}{\kappa + 2}\right) \quad E_0 R^3 \quad , \quad b_2 = -\left(\frac{3}{\kappa + 2}\right) E_0$$

(b) Inserting the values of b_1 and b_2 in Eq. (1), we get

$$\mathbf{E} = \begin{cases} E_0 \hat{\mathbf{z}} + E_0 \left(\dfrac{\kappa - 1}{\kappa + 2}\dfrac{R^3}{r^3}\right)\left(2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\theta}\right), & \text{for } r > R \\ \dfrac{3}{\kappa + 2} E_0 \hat{\mathbf{z}}, & \text{for } r < R \end{cases}$$

(c) The polarization is given by $\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E}$ and we find a non-zero answer for $r < R$. We obtain

$$\mathbf{P} = \frac{3(\kappa - 1)}{\kappa + 2} \epsilon_0 E_0 \hat{\mathbf{z}}$$

which is (somewhat surprisingly) a constant vector. The total dipole moment of the sphere is $4\pi R^3/3$ times the constant value of $\mathbf{P}$ given above.

(d) The volume and bound charge densities can be computed easily using the standard formulae:

$$\rho_b(\theta, \varphi) = -\nabla \mathbf{P} = 0, \sigma_b(\theta, \varphi) = \mathbf{P} \cdot \hat{n} = \mathbf{P} \cdot \hat{\mathbf{r}} = \frac{3(\kappa - 1)}{\kappa + 2} \epsilon_0 E_0 \cos\theta$$

8. Problem-8

Question: A cylindrical coaxial cable, lying along the z-axis has conducting surfaces at $\rho = a$ and $\rho = 4a$, which carry uniform surface charge densities σ_0 and $-\sigma_0/4$, respectively. The region between the two conductors is filled with linear dielectric media with dielectric constants K_1 and K_2 in the regions $a < \rho \leq 2a$ and $2a < \rho < 4a$, respectively.

$D = \epsilon_0 E + P$
$D(1-1/K_1) = P$

$E = D/\epsilon_0 K_1$

$\frac{1}{2} \cdot \frac{\sigma_0^2 a^2}{\epsilon_0 K_1}$

(a) Find the energy density between $a < \rho \leq 2a$.

(b) Determine the ratio of the magnitude of the polarization just inside and just outside the boundary at $\rho = 2a$.

$\frac{1-1/K_1}{1-1/K_2}$

(c) Sketch $|E|$ as a function of ρ in the interval $0 < \rho \leq 5a$ when $K_1 = 1.5$ and $K_2 = 2$.

Solution:

We use the Gauss's law for dielectric to evaluate D in the various regions of interest: For $0 < \rho < a$, we see that Q_f the free charge enclosed is zero. Thus, $D = 0$ and hence $E = 0$. In similar fashion, $\rho > 4a$, the enclosed charge is once again zero thus once again $D = 0$, and $E = 0$ in this region. For $a < \rho < 4a, Q_f^{\text{enc}} = 2\pi\sigma_0 aL$. Thus by using Gauss's law for dielectrics we get

$$\int_S \mathbf{D} \cdot d\mathbf{S} = Q_f^{\text{enc}}$$

Using a cylindrical Gaussian surface, one sees that

$$D_\rho = \frac{\sigma_0 a}{\rho}.$$

(a) To evaluate the energy density in the region $a < \rho < 2a$, we use Energy density $= \frac{1}{2}\mathbf{D} \cdot \mathbf{E}$, with $\mathbf{E}(a < \rho < 2a) = \frac{\sigma_0 a}{\epsilon_0 K_1 \rho}$. Thus, the energy density

$$W = \frac{1}{2} \frac{\sigma_0^2 a^2}{\epsilon_0 K_1 \rho^2}$$

(b) We have

$$P_\rho = D_\rho - \epsilon_0 E_\rho$$

Thus, for $0 < \rho < 2a$, we have

$$P_\rho = (1 - 1/K_1) \frac{\sigma_0 a}{\rho}$$

and for $\rho > 2a$, we get

$$P_\rho = (1 - 1/K_2) \frac{\sigma_0 a}{\rho}$$

The ratio of polarization at the boundary is thus

$$= \frac{K_1 - 1}{K_1} \frac{K_2}{K_2 - 1}.$$

(c) From above we see that $|\mathbf{E}(a < \rho < 2a)| = \frac{\sigma_0 a}{\epsilon_0 K_1 \rho}$, and $|\mathbf{E}(2a < \rho < 4a)| = \frac{\sigma_0 a}{\epsilon_0 K_2 \rho}$.

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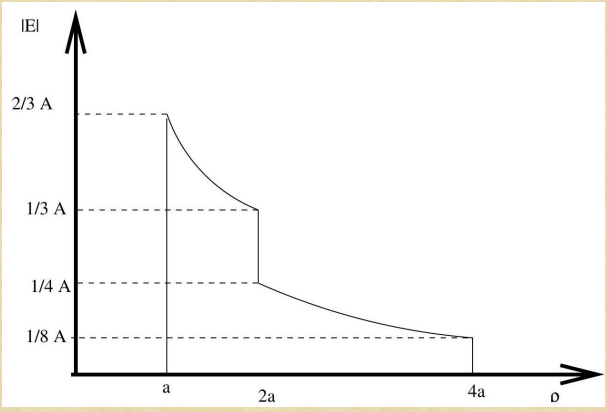


Figure 5: A plot of $E(\rho)$ vs ρ : The discontinuities of the Electric field across the various boundaries are shown. The constant $A = \sigma_0/\epsilon_0$.